

Andrew Estrada

andrewgestrada@gmail.com ❖ andrewgestrada.com

EDUCATION

Northern Arizona University

B.S, Mechanical Engineering

Aug. 2020 – May 2024

Flagstaff, AZ

- NAU Founders Gold Scholarship
- Member of, Formula SAE, SAE Baja, Space Jacks, and ASME

WORK EXPERIENCE

Parker-Meggitt- US Safety Systems

July 2023 – Aug. 2023

Manufacturing Engineering Intern

Simi Valley, CA

- Designed and implemented innovative fixtures to enhance process safety and ergonomics, resulting in increased efficiency, productivity, and adherence to LEAN manufacturing
- Collaborated with design engineers to conduct failure analyses and implement effective corrective actions
- Provided invaluable assistance to production personnel by updating and clarifying work instructions, optimizing production processes and ensuring smooth operations

Accu-Glass Products, Inc.

May 2022 – Aug. 2022

Engineering Intern

Santa Clarita, CA

- Responsible for operating Haas CNC Mills and Lathes to complete components for capital projects
- Constructed 3D CAD models and technical drawings for manufacture
- Aided in process for work holding and manufacturing methods
- Initiated precision quality control and effectively took precise measurements to ensure effective tolerance for high-accuracy air to high and ultra-high vacuum parts

RELEVANT PROJECTS

- **Formula SAE: Northern Arizona University** **April 2023 – May 2024**
 - Front Pushrod Suspension Lead, responsible for design, validated geometry within LOTUS Suspension Shark, utilized ANSYS to simulate driving conditions, collaboration with chassis team for successful integration
 - Manufactured Front Pushrod Assembly utilizing CNC Mill and Lathe, programming with Mastercam
 - Team Test Engineer responsible for test planning, data acquisition and analysis (DAQ), evaluating design choices and vehicle dynamics against lap simulations to suggest car setup changes.
 - Key asset in founding a Formula SAE team at Northern Arizona University for spring 2024 competition
 - Sponsor relations manager, new member outreach coordinator, website developer and social media manager
- **Pressure Gauge Testing Fixture: Parker-Meggitt** **Summer 2023**
 - Collaborated with production floor to enhance process safety and compliance of pressure gauge testing procedure
 - Designed innovative testing fixture capable of cycling multiple gauges simultaneously, including the creation of detailed CAD models and assembly drawings.
 - Engineered a safety shield to protect personnel during testing processes, prioritizing safety standards.
 - Worked closely with Tool Design Engineers to validate the fixture's design and facilitate its fabrication.
- **Eagle Scout Project: Boy Scouts of America** **Fall 2019**
 - Conceptualized community service project in partnership with the city of Santa Clarita, California, over a course of one year, fundraised and presented to local businesses and led 50+ volunteers in the renovation and beautification of Haskell Canyon Open Space
 - Created technical drawings for the trail kiosk and trail signs utilizing Autodesk Inventor

SKILLS & INTERESTS

- **Technical Skills:** SolidWorks; ANSYS; Fusion 360; Inventor; MATLAB; Microsoft Office Suite; Mastercam;
- **Manufacturing Skills:** Haas 3-Axis CNC Mill and 2 & 4-Axis Lathe Experience; FDM 3D Printing; Understanding of Lean and Six Sigma Manufacturing Methodology
- **Interests:** Mountain Biking; High Power Rocketry; Sound Systems and Acoustics; Running; Backpacking; Travel